Solutions to Dummit & Foote's Abstract Algebra 3rd Edition

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Preface

This is a collection of solutions to the exercises in Dummit and Foote's *Abstract Algebra*, 3rd edition. These solutions were written by me as I worked through the book, and are intended to serve as a reference for myself and others who are studying abstract algebra. I have made every effort to ensure the correctness of these solutions, but I cannot guarantee that they are free of errors. If you find any mistakes or have suggestions for improvement, please feel free to contact me.

I've attempted to format the solutions in a clear and consistent manner, using LaTeX for typesetting. Each chapter's exercises are included in separate files for better organization if viewing on GitHub. Moreover, solutions to exercises only utilize techniques that are available to an advanced undergraduate student or beginning graduate student, in line with the intended audience of the textbook. As an aside, I've graduated from a bachelor's program in mathematics but have not pursued graduate studies, so my background is primarily at the undergraduate level.

Some suggestions I've been given to improve the guide are citing particularly important exercises that will prove useful in reading subsequent chapters, and providing writeups/discussions for why I choose a particular method of solution (such as functions/homomorphisms that may appear out of nowhere). Most problems are enclosed in the following environment:

Exercise 0.0.1

Example exercise.

However, there are some exercises that I found to be either challenging, interesting, or useful for the development of later material. These exercises are enclosed in a special environment:

($*$) **Exercise 0.0.2**

Example special exercise.

This is to highlight these exercises for future readers who may want to focus on them.

Many thanks to the authors, David S. Dummit and Richard M. Foote, for writing such an excellent textbook that has been a valuable resource for my recreational studies in abstract algebra. I've received requests on Reddit for individuals to assist me in completing this project, but at this time I prefer to work on it independently. However, I welcome feedback and suggestions from anyone who is interested in contributing to the project in the future.

5 Direct and Semidirect Products and Abelian Groups

5.1 Direct Products

Exercise 5.1.1

Show that the center of a direct product is the direct product of the centers:

$$Z(G_1 \times G_2 \times \cdots \times G_n) = Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n)$$

Deduce that the direct product of two groups is abelian if and only if each of the factors is abelian.

Solution. Let $G = G_1 \times G_2 \times \cdots \times G_n$. Let $z = (z_1, z_2, \ldots, z_n) \in Z(G)$. Then for any $g = (g_1, g_2, \ldots, g_n) \in G$, we have $zg = gz$, or equivalently,

$$(z_1 g_1, z_2 g_2, \ldots, z_n g_n) = (g_1 z_1, g_2 z_2, \ldots, g_n z_n)$$

This implies that $z_i g_i = g_i z_i$ for all i, so $z_i \in Z(G_i)$ for all i. Thus, $z \in Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n)$, and we have shown that $Z(G) \subseteq Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n)$.

Conversely, let $z = (z_1, z_2, \ldots, z_n) \in Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n)$. Then for any $g = (g_1, g_2, \ldots, g_n) \in G$, we have

$$zg = (z_1 g_1, z_2 g_2, \ldots, z_n g_n) = (g_1 z_1, g_2 z_2, \ldots, g_n z_n) = gz$$

Thus, $z \in Z(G)$, and we have shown that $Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n) \subseteq Z(G)$. ■

Exercise 5.1.2

Let $G_1, G_2, \ldots, G_n$ be groups and $G = G_1 \times \cdots \times G_n$. Let I be a proper, nonempty subset of $\{1, \ldots, n\}$ and let $J = \{1, \ldots, n\} - I$. Define G_I to be the set of elements of G that have the identity G_j in position j for all $j \in J$.

(a) Prove that G_I is isomorphic to the direct product of the G_i for $i \in I$.

(b) Prove that G_I is a normal subgroup of G and that $G/G_I \cong G_J$.

(c) Prove that $G \cong G_I \times G_J$.

Solution.

(a) Identifying G_i with the subgroup of G defined in Proposition 5.2, let H be the product of the G_i for $i \in I$ in G. Let $\varphi : G_I \to H$ be the map defined by mapping the non-identity coordinates of G_I to the corresponding coordinates of H. It is clear that φ is a homomorphism. Moreover, φ is surjective by definition of H. Finally, the kernel of φ is the set of all elements of G_I that have the identity in all coordinates, which is just the identity element of G. Thus, φ is an isomorphism from G_I to H, and we are done.

(b) Consider the map $\varphi : G \to G_J$ defined by mapping the non-identity coordinates of G to the corresponding coordinates of G_J. It is clear that φ is a homomorphism. Moreover, φ is surjective by definition of G_J. Finally, the kernel of φ is the set of all elements of G that have the identity in all coordinates corresponding to J, which is exactly G_I. Thus, φ induces an isomorphism from G/G_I to G_J, and we are done.

(c) Any element of $G_I \times G_J$ can be written as the product of an element of G_I and an element of G_J by taking the first and second coordinates, respectively. Thus, the map $\varphi : G_I \times G_J \to G$ defined by $\varphi((g_i)_{i \in I}, (g_j)_{j \in J}) = (g_1, g_2, \ldots, g_n)$ is a bijection. Injectivity is clear, since $\ker \varphi$ is trivial. It is also a homomorphism since the group operation in G is defined coordinate-wise. Thus, φ is an isomorphism from $G_I \times G_J$ to G, and we are done. ■

Exercise 5.1.3

Under the notation of the preceding exercise, let I and K be any disjoint nonempty subsets of $\{1, 2, \ldots, n\}$, and let G_I and G_K be the subgroups of G defined above. Prove that $xy = yx$ for all $x \in G_I$ and $y \in G_K$.

Solution. Let $x \in G_I$ and $y \in G_K$. Then x has the identity in all coordinates corresponding to $J = \{1, 2, \ldots, n\} - I$, and y has the identity in all coordinates corresponding to $L = \{1, 2, \ldots, n\} - K$. Since I and K are disjoint, we have $J \cup L = \{1, 2, \ldots, n\}$, so every coordinate of x and y is the identity in at least one of them. Thus, when we compute the product xy, we have

$$xy = (x_1y_1, x_2y_2, \ldots, x_ny_n) = (y_1x_1, y_2x_2, \ldots, y_nx_n) = yx$$

as desired. ■

Exercise 5.1.4

Let A and B be finite groups, and let p be a prime. Prove that any Sylow p-subgroup of $A \times B$ is of the form $P \times Q$, where $P \in \mathrm{Syl}_p(A)$ and $Q \in \mathrm{Syl}_p(B)$. Prove that $n_p(A \times B) = n_p(A)n_p(B)$. Generalize both of these results to a direct product of any finite number of finite groups (so that the number of Sylow p-subgroups of a direct product is the product of the numbers of Sylow p-subgroups of the factors).

Solution. Let $|A| = p^\alpha m$ and $|B| = p^\beta n$, where $p \nmid m$ and $p \nmid n$. Then $|A \times B| = |A||B| = p^{\alpha+\beta}mn$. Let $R \in \mathrm{Syl}_p(A \times B)$. Then $|R| = p^{\alpha+\beta}$, so R is a Sylow p-subgroup of $A \times B$. Let $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$ be the projection maps. Then $\pi_A(R)$ is a p-subgroup of A, so $\pi_A(R) \subseteq P$ for some $P \in \mathrm{Syl}_p(A)$. Similarly, $\pi_B(R) \subseteq Q$ for some $Q \in \mathrm{Syl}_p(B)$. Thus, $R \subseteq P \times Q$. Since $|P \times Q| = |P||Q| = p^{\alpha+\beta}$, we must have $R = P \times Q$.

To show $n_p(A \times B) = n_p(A)n_p(B)$, we note that the Sylow p-subgroups of $A \times B$ are precisely the subgroups of the form $P \times Q$, where $P \in \mathrm{Syl}_p(A)$ and $Q \in \mathrm{Syl}_p(B)$. Since there are $n_p(A)$ choices for P and $n_p(B)$ choices for Q, we have $n_p(A \times B) = n_p(A)n_p(B)$. ■

Exercise 5.1.5

Exhibit a nonnormal subgroup of $Q_8 \times Z_4$ (note that every subgroup of each factor is normal).

Solution. Note that Q_8 is non-abelian, so we may use the idea of a diagonal subgroup from Exercise 3.1.38. In particular, let $H = \langle (i, 1) \rangle = \{(1, 0), (i, 1), (-1, 2), (-i, 3)\}$. Then H is a subgroup of $Q_8 \times Z_4$ that is isomorphic to $\langle i \rangle$. However,

$$(j, 0)(i, 1)(j, 0)^{-1} = (ji(-j), 0 + 1 + 0) = (-i, 1)$$

is not in H, so H is not normal in $Q_8 \times Z_4$. ■

Exercise 5.1.6

Show that all subgroups of $Q_8 \times E_{2^n}$ are normal.

Solution. Let $H \leq Q_8 \times E_{2^n}$. Suppose $(a, b) \in H$ and $(x, y) \in Q_8 \times E_{2^n}$. Note that $(a^{-1}, b) \in H$, since every $b \in E_{2^n}$ is its own inverse. Then

$$(x, y)(a, b)(x, y)^{-1} = (xax^{-1}, yby^{-1}) = (xax^{-1}, b)$$

since E_{2^n} is abelian. If $a \in Z(Q_8)$, then $xax^{-1} = a$, so $(x, y)(a, b)(x, y)^{-1} = (a, b) \in H$. If $a \notin Z(Q_8)$, then a is one of i, j, k or their inverses. In all cases, we have xax^{-1} is one of i, j, k or their inverses, so $xax^{-1} \in H$. Thus, $(x, y)(a, b)(x, y)^{-1} \in H$ for all $(a, b) \in H$ and $(x, y) \in Q_8 \times E_{2^n}$, so H is normal in $Q_8 \times E_{2^n}$. ■

Exercise 5.1.7

Let $G_1, G_2, \ldots, G_n$ be groups and π be a fixed element of S_n. Prove that the map

$$\varphi_\pi : G_1 \times G_2 \times \cdots \times G_n \to G_{\pi^{-1}(1)} \times G_{\pi^{-1}(2)} \times \cdots \times G_{\pi^{-1}(n)}$$

defined by

$$\varphi_\pi(g_1, g_2, \ldots, g_n) = (g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \ldots, g_{\pi^{-1}(n)})$$

is an isomorphism (so that changing the order of the factors in a direct product does not change the isomorphism type).

Solution. Let $\pi \in S_n$ and let φ_π be the map defined in the problem statement. Let $g = (g_1, g_2, \ldots, g_n)$ and $h = (h_1, h_2, \ldots, h_n)$ be elements of $G_1 \times G_2 \times \cdots \times G_n$. Then

$$\begin{aligned}
\varphi_\pi(gh) &= \varphi_\pi(g_1h_1, g_2h_2, \ldots, g_nh_n) \\
&= (g_{\pi^{-1}(1)}h_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}h_{\pi^{-1}(2)}, \ldots, g_{\pi^{-1}(n)}h_{\pi^{-1}(n)}) \\
&= (g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \ldots, g_{\pi^{-1}(n)})(h_{\pi^{-1}(1)}, h_{\pi^{-1}(2)}, \ldots, h_{\pi^{-1}(n)}) \\
&= \varphi_\pi(g)\varphi_\pi(h)
\end{aligned}$$

Noting that $\pi \in S_n$ is a bijection, consider the map $\varphi_{\pi^{-1}} : G_{\pi^{-1}(1)} \times G_{\pi^{-1}(2)} \times \cdots \times G_{\pi^{-1}(n)} \to G_1 \times G_2 \times \cdots \times G_n$ defined by

$$\varphi_{\pi^{-1}}(g_1, g_2, \ldots, g_n) = (g_{\pi(1)}, g_{\pi(2)}, \ldots, g_{\pi(n)})$$

Then $\varphi_{\pi^{-1}}$ is a homomorphism by the same argument as above. Moreover, $\varphi_{\pi^{-1}}$ is the inverse of φ_π, since

$$\varphi_\pi(\varphi_{\pi^{-1}}(g_1, g_2, \ldots, g_n)) = \varphi_\pi(g_{\pi(1)}, g_{\pi(2)}, \ldots, g_{\pi(n)}) = (g_1, g_2, \ldots, g_n)$$

and

$$\varphi_{\pi^{-1}}(\varphi_\pi(g_1, g_2, \ldots, g_n)) = \varphi_{\pi^{-1}}(g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \ldots, g_{\pi^{-1}(n)}) = (g_1, g_2, \ldots, g_n).$$

Therefore, φ_π is an isomorphism. ■

Exercise 5.1.8

Let $G_1 = G_2 = \cdots = G_n$ and let $G = G_1 \times \cdots \times G_n$. Under the notation of the preceding exercise show that $\varphi_\pi \in \mathrm{Aut}(G)$. Show also that the map $\pi \mapsto \varphi_\pi$ is an injective homomorphism of S_n into $\mathrm{Aut}(G)$. (In particular, $\varphi_{\pi_1} \circ \varphi_{\pi_2} = \varphi_{\pi_1\pi_2}$. It is at this point that the π^{-1}'s in the definition of φ_π are needed. The underlying reason for this is because if e_i is the n-tuple with 1 in position i and zeros elsewhere, $1 \leq i \leq n$, then S_n acts on $\{e_1, \ldots, e_n\}$ by $\pi \cdot e_i = e_{\pi(i)}$; this is a left group action. If the n-tuple $(g_1, \ldots, g_n)$ is represented by $g_1e_1 + \cdots + g_ne_n$, then this left group action on $\{e_1, \ldots, e_n\}$ extends to a left group action on sums by

$$\pi \cdot (g_1e_1 + g_2e_2 + \cdots + g_ne_n) = g_1e_{\pi(1)} + g_2e_{\pi(2)} + \cdots + g_ne_{\pi(n)}.$$

The coefficient of $e_{\pi(i)}$ on the right hand side is g_i, so the coefficient of e_i is $g_{\pi^{-1}(i)}$. Thus the right hand side may be rewritten as $g_{\pi^{-1}(1)}e_1 + g_{\pi^{-1}(2)}e_2 + \cdots + g_{\pi^{-1}(n)}e_n$, which is precisely the sum attached to the n-tuple $(g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \ldots, g_{\pi^{-1}(n)})$. In other words, any permutation of the "position vectors" $e_1, \ldots, e_n$ (which fixes their coefficients) is the same as the inverse permutation on the coefficients (fixing the e_i's). If one uses π's in place of π^{-1}'s in the definition of φ_π then the map $\pi \mapsto \varphi_\pi$ is not necessarily a homomorphism — it corresponds to a *right* group action.)

Solution. Note that $\varphi_\pi \in \mathrm{Aut}(G)$ since φ_π is an isomorphism from G to itself. Let $\pi_1, \pi_2 \in S_n$. Then

$$\begin{aligned}
\varphi_{\pi_1}(\varphi_{\pi_2}(g_1, g_2, \ldots, g_n)) &= \varphi_{\pi_1}(g_{\pi_2^{-1}(1)}, g_{\pi_2^{-1}(2)}, \ldots, g_{\pi_2^{-1}(n)}) \\
&= (g_{\pi_2^{-1}(\pi_1^{-1}(1))}, g_{\pi_2^{-1}(\pi_1^{-1}(2))}, \ldots, g_{\pi_2^{-1}(\pi_1^{-1}(n))}) \\
&= (g_{(\pi_1\pi_2)^{-1}(1)}, g_{(\pi_1\pi_2)^{-1}(2)}, \ldots, g_{(\pi_1\pi_2)^{-1}(n)}) \\
&= \varphi_{\pi_1\pi_2}(g_1, g_2, \ldots, g_n)
\end{aligned}$$

Therefore, the map $\pi \mapsto \varphi_\pi$ is a homomorphism. To see that it is injective, note that if φ_π is the identity automorphism, then for all i, $g_{\pi^{-1}(i)} = g_i$ for all $g_i \in G_i$, which implies π is the identity permutation. Hence, the map is injective. ■

Exercise 5.1.9

Let G_i be a field F for all i and use the preceding exercise to show that the set of $n \times n$ matrices with one 1 in each row and each column is a subgroup of $\mathrm{GL}_n(F)$ isomorphic to S_n (these matrices are the *permutation matrices* since they simply permute the standard basis $e_1, \dots, e_n$ (as above) of the n-dimensional vector space $F \times F \times \cdots \times F$).

Solution. Let $G = F \times F \times \cdots \times F$ (n times) and let $\pi \in S_n$. Then φ_π is an automorphism of G that permutes the standard basis vectors $e_1, e_2, \dots, e_n$. In particular, $\varphi_\pi(e_i) = e_{\pi(i)}$ for all i. Thus, the matrix representation of φ_π with respect to the standard basis is a permutation matrix. Since the map $\pi \mapsto \varphi_\pi$ is an injective homomorphism from S_n to $\mathrm{Aut}(G)$, the set of permutation matrices forms a subgroup of $\mathrm{GL}_n(F)$ that is isomorphic to S_n. ■

Exercise 5.1.10

Let p be a prime. Let A and B be two cyclic groups of order p with generators x and y respectively. Set $E = A \times B$ so that E is the elementary abelian group of order p^2: E_{p^2} Prove that the distinct subgroups of E of order p are

$$\langle x \rangle, \quad \langle xy \rangle, \quad \langle xy^2 \rangle, \quad \dots, \quad \langle xy^{p-1} \rangle, \quad \langle y \rangle$$

(note that there are $p + 1$ of them).

Solution. Note that any proper, nontrivial subgroup of E must have order p, and every element of E has order p. The text shows that there are $p+1$ subgroups of E of order p, so it suffices to show that the subgroups listed in the problem statement are distinct. Note that $\langle x \rangle$ and $\langle y \rangle$ are distinct since they have different generators. Moreover, $\langle xy^i \rangle$ is distinct from $\langle x \rangle$ and $\langle y \rangle$ for all i, since xy^i is not a power of x or y (as can be seen because $\langle x \rangle \cap \langle y \rangle$ is trivial). Finally, $\langle xy^i \rangle$ is distinct from $\langle xy^j \rangle$ for $i \neq j$, since if $\langle xy^i \rangle = \langle xy^j \rangle$, then xy^i is a power of xy^j, which implies y^{i-j} is a power of x, which is impossible since $\langle x \rangle \cap \langle y \rangle$ is trivial. Thus, all the subgroups listed in the problem statement are distinct. ■

Exercise 5.1.11

Let p be a prime, and let $n \in \mathbb{Z}^+$. Find a formula for the number of subgroups of order p in the elementary abelian group E_{p^n}.

Solution. Every element of E_{p^n} has order p, so every subgroup of order p is generated by a single non-identity element. Since there are $p^n - 1$ non-identity elements in E_{p^n}, and each subgroup of order p has $p - 1$ generators (the non-identity elements of the subgroup), the number of subgroups of order p in E_{p^n} is

$$\frac{p^n - 1}{p - 1} = 1 + p + p^2 + \cdots + p^{n-1}$$

as desired. ■

(∗) Exercise 5.1.12

Let A and B be groups. Assume $Z(A)$ contains a subgroup Z_1 and $Z(B)$ contains a subgroup Z_2 with $Z_1 \cong Z_2$. Let this isomorphism be given by the map $x_i \mapsto y_i$ for all $x_i \in Z_1$. A ***central product*** of A and B is a quotient

$$(A \times B)/Z \quad \text{where} \quad Z = \{(x_i, y_i^{-1}) \mid x_i \in Z_1\}$$

and is denoted by $A * B$ — it is not unique since it depends on Z_1, Z_2 and the isomorphism between them. (Think of $A * B$ as the direct product of A and B "collapsed" by identifying each element $x_i \in Z_1$ with its corresponding element $y_i \in Z_2$.)

(a) Prove that the images of A and B in the quotient group $A * B$ are isomorphic to A and B, respectively, and that these images intersect in a central subgroup isomorphic to Z_1. Find $|A * B|$.

(b) Let $Z_4 = \langle x \rangle$. Let $D_8 = \langle r, s \rangle$ and $Q_8 = \langle i, j \rangle$ be given by their usual generators and relations. Let $Z_4 * D_8$ be the central product of Z_4 and D_8 which identifies x^2 and r^2 (i.e., $Z_1 = \langle x^2 \rangle$, $Z_2 = \langle r^2 \rangle$ and the isomorphism is $x^2 \mapsto r^2$) and let $Z_4 * Q_8$ be the central product of Z_4 and Q_8 which identifies x^2 and -1. Prove that $Z_4 * D_8 \cong Z_4 * Q_8$.

Solution.

(a) Consider the maps $\pi_A : A \to A * B$ and $\pi_B : B \to A * B$ defined by $\pi_A(a) = (a, 1)Z$ and $\pi_B(b) = (1, b)Z$. It is clear that π_A and π_B are homomorphisms. Moreover, $\ker \pi_A$ consists of the elements $a \in A$ such that $(a, 1) \in Z$. Then $y_i = 1$, and since $x_i \mapsto y_i$ is an isomorphism, we have $x_i = 1$, so $a = 1$. Thus, $\ker \pi_A$ is trivial, and π_A is injective. Similarly, $\ker \pi_B$ is trivial, and π_B is injective. Therefore, the images of A and B in $A * B$ are isomorphic to A and B, respectively. Moreover, the images of A and B in $A * B$ intersect in the subgroup $\{(x_i, 1)Z \mid x_i \in Z_1\}$, which is isomorphic to Z_1. Finally, since Z has $|Z_1|$ elements, we have

$$|A * B| = \frac{|A||B|}{|Z_1|}.$$

(b) Let Z_D denote the Z in the quotient for $Z_4 * D_8$ and let Z_Q denote the Z in the quotient for $Z_4 * Q_8$. Let x generate Z_4 in $Z_4 * D_8$ and y generate Z_4 in $Z_4 * Q_8$. Let r and s be the generators of D_8 in $Z_4 * D_8$ and let i and j be the generators of Q_8 in $Z_4 * Q_8$. Let $\mathbf{x}$ denote the coset $(x, 1)Z_D$ in $Z_4 * D_8$, and use similar notation for the elements r, s, y, i, j. Let $\mathbf{z} = \mathbf{j}\mathbf{y}^{-1}$. Consider the map $\varphi : Z_4 * D_8 \to Z_4 * Q_8$ defined by

$$\varphi(\mathbf{x}^\alpha \mathbf{r}^\beta \mathbf{s}^\gamma) = \mathbf{y}^\alpha \mathbf{i}^\beta \mathbf{z}^\gamma.$$

To see that this is well-defined, suppose $\mathbf{x}^\alpha \mathbf{r}^\beta \mathbf{s}^\gamma = \mathbf{1}$ in $Z_4 * D_8$. Then $\mathbf{r}^\beta \mathbf{s}^\gamma$ is in the image of Z_4 in $Z_4 * D_8$, so $\mathbf{r}^\beta \mathbf{s}^\gamma = \mathbf{x}^{2\delta}$ for some δ. Since r^2 is central in D_8, we have $\mathbf{r}^\beta \mathbf{s}^\gamma = \mathbf{r}^{\beta+2\delta} \mathbf{s}^\gamma$. If $\gamma = 0$, then $\mathbf{r}^{\beta+2\delta} = \mathbf{1}$, so $4 \mid \beta + 2\delta$. If $\gamma = 1$, then $\mathbf{r}^{\beta+2\delta}\mathbf{s} = \mathbf{1}$, so $4 \mid \beta + 2\delta - 2$. In either case, we have $4 \mid \beta + 2\delta$, so $\mathbf{r}^\beta \mathbf{s}^\gamma = \mathbf{x}^{2\delta} = \mathbf{1}$. Thus, α is even, and we have $\varphi(\mathbf{x}^\alpha \mathbf{r}^\beta \mathbf{s}^\gamma) = \mathbf{y}^\alpha \mathbf{i}^\beta \mathbf{z}^\gamma = \mathbf{1}$. Therefore, φ is well-defined. It is clear that φ is a homomorphism. Moreover, if $\varphi(\mathbf{x}^\alpha \mathbf{r}^\beta \mathbf{s}^\gamma) = \mathbf{1}$, then α is even and β and γ are zero, so φ is injective. Finally, since $|Z_4 * D_8| = |Z_4 * Q_8|$, we conclude that φ is an isomorphism. ∎

Exercise 5.1.13

Give presentations for the groups $Z_4 * D_8$ and $Z_4 * Q_8$ constructed in the preceding exercise.

Solution. To construct the presentations, we note that $\mathbf{x}$ should satisfy the relations of Z_4, $\mathbf{r}$ and $\mathbf{s}$ should satisfy the relations of D_8, and $\mathbf{i}$ and $\mathbf{z}$ should satisfy the relations of Q_8. Moreover, we have the relations $\mathbf{x}^2 = \mathbf{r}^2$ in $Z_4 * D_8$, since $\mathbf{x}^2\mathbf{r}^{-2} = \mathbf{1}$. Lastly, $\mathbf{x}$ is central in that $\mathbf{xr} = \mathbf{rx}$ and $\mathbf{xs} = \mathbf{sx}$ in $Z_4 * D_8$. Then a presentation for $Z_4 * D_8$ is

$$\langle \mathbf{x}, \mathbf{r}, \mathbf{s} \mid \mathbf{r}^4 = \mathbf{s}^2 = \mathbf{1}, \mathbf{x}^2 = \mathbf{r}^2, \mathbf{srs} = \mathbf{r}^{-1}, \mathbf{xr} = \mathbf{rx}, \mathbf{xs} = \mathbf{sx} \rangle.$$

To find a presentation for $Z_4 * Q_8$, we may use the isomorphism φ from the preceding exercise to rewrite the presentation for $Z_4 * D_8$ in terms of $\mathbf{y}, \mathbf{i}, \mathbf{z}$. In particular, we have $\mathbf{y}^2 = \mathbf{i}^2\mathbf{z}^2$, $\mathbf{z}^2 = \mathbf{1}$, and $\mathbf{yi} = \mathbf{iyz}$. Thus, a presentation for $Z_4 * Q_8$ is

$$\langle \mathbf{y}, \mathbf{i}, \mathbf{z} \mid \mathbf{i}^4 = \mathbf{z}^2 = \mathbf{1}, \mathbf{y}^2 = \mathbf{i}^2, \mathbf{ziz} = \mathbf{i}^{-1}, \mathbf{yi} = \mathbf{iy}, \mathbf{yz} = \mathbf{zy} \rangle.$$ ∎

Exercise 5.1.14

Let $G = A_1 \times A_2 \times \cdots \times A_n$, and for each i, let B_i be a normal subgroup of A_i. Prove that $B_1 \times B_2 \times \cdots \times B_n \trianglelefteq G$, and that

$$(A_1 \times A_2 \times \cdots \times A_n)/(B_1 \times B_2 \times \cdots \times B_n) \cong (A_1/B_1) \times (A_2/B_2) \times \cdots \times (A_n/B_n).$$

Solution. Let $G = A_1 \times A_2 \times \cdots \times A_n$ and let $B_i \trianglelefteq A_i$ for all i. Let $H = B_1 \times B_2 \times \cdots \times B_n$. It is clear that H is a subgroup of G. Moreover, since each B_i is normal in A_i, we have that for any $g = (a_1, a_2, \ldots, a_n) \in G$ and any $h = (b_1, b_2, \ldots, b_n) \in H$, we have

$$ghg^{-1} = (a_1 b_1 a_1^{-1}, a_2 b_2 a_2^{-1}, \ldots, a_n b_n a_n^{-1}) \in H$$

since $a_i b_i a_i^{-1} \in B_i$ for all i. Thus, H is normal in G.

Consider the map $\varphi : G \to (A_1/B_1) \times (A_2/B_2) \times \cdots \times (A_n/B_n)$ defined by

$$\varphi((a_1, a_2, \ldots, a_n)) = (a_1 B_1, a_2 B_2, \ldots, a_n B_n).$$

It is clear that φ is a homomorphism. Moreover, φ is surjective since for any $(a_1 B_1, a_2 B_2, \ldots, a_n B_n)$, we have $\varphi((a_1, a_2, \ldots, a_n)) = (a_1 B_1, a_2 B_2, \ldots, a_n B_n)$. Finally, the kernel of φ is the set of all $(a_1, a_2, \ldots, a_n)$ such that $a_i \in B_i$ for all i, which is exactly H. Thus, by the First Isomorphism Theorem, we have

$$G/H \cong (A_1/B_1) \times (A_2/B_2) \times \cdots \times (A_n/B_n)$$

as desired. ■

Exercise 5.1.15

Let I be any nonempty index set and let $(G_i, \star_i)$ be a group for each $i \in I$. The *direct product* of the groups $G_i, i \in I$ is the set $G = \prod_{i \in I} G_i$ (the Cartesian product of the G_i's) with a binary operation defined as follows: if $\prod a_i$ and $\prod b_i$ are elements of G, then their product in G is given by

$$\left(\prod_{i \in I} a_i\right)\left(\prod_{i \in I} b_i\right) = \prod_{i \in I} (a_i \star_i b_i)$$

(i.e., the group operation in the direct product is defined componentwise).

(a) Show that this binary operation is well defined and associative.

(b) Show that the element $\prod 1_i$ satisfies the axiom for the identity of G, where 1_i is the identity of G_i for all i.

(c) Show that the element $\prod a_i^{-1}$ is the inverse of $\prod a_i$, where the inverse of each component element a_i is taken in the group G_i.

Conclude that the direct product is a group. (Note that if $I = \{1, 2, \ldots, n\}$, this definition of the direct product is the same as the n-tuple definition in the text.)

Solution.

(a) To show well-definedness, let $\prod \alpha_i = \prod a_i$ and $\prod \beta_i = \prod b_i$. Then $\alpha_i = a_i$ and $\beta_i = b_i$ for all i, so

$$\left(\prod_{i \in I} \alpha_i\right)\left(\prod_{i \in I} \beta_i\right) = \prod_{i \in I} (\alpha_i \star_i \beta_i) = \prod_{i \in I} (a_i \star_i b_i)$$

as desired. To show associativity, let $\prod a_i$, $\prod b_i$, and $\prod c_i$ be elements of G. Then

$$\begin{aligned}\left(\prod_{i\in I} a_i\right)\left(\left(\prod_{i\in I} b_i\right)\left(\prod_{i\in I} c_i\right)\right) &= \left(\prod_{i\in I} a_i\right)\left(\prod_{i\in I}(b_i \star_i c_i)\right)\\ &= \prod_{i\in I}(a_i \star_i (b_i \star_i c_i))\\ &= \prod_{i\in I}((a_i \star_i b_i) \star_i c_i)\\ &= \left(\prod_{i\in I}(a_i \star_i b_i)\right)\left(\prod_{i\in I} c_i\right)\\ &= \left(\left(\prod_{i\in I} a_i\right)\left(\prod_{i\in I} b_i\right)\right)\left(\prod_{i\in I} c_i\right)\end{aligned}$$

as desired.

(b) Let $\prod a_i$ be an element of G. Then

$$\left(\prod_{i\in I} 1_i\right)\left(\prod_{i\in I} a_i\right) = \prod_{i\in I}(1_i \star_i a_i) = \prod_{i\in I} a_i$$

and

$$\left(\prod_{i\in I} a_i\right)\left(\prod_{i\in I} 1_i\right) = \prod_{i\in I}(a_i \star_i 1_i) = \prod_{i\in I} a_i.$$

Thus, $\prod 1_i$ is the identity element of G.

(c) Let $\prod a_i$ be an element of G. Then

$$\left(\prod_{i\in I} a_i\right)\left(\prod_{i\in I} a_i^{-1}\right) = \prod_{i\in I}(a_i \star_i a_i^{-1}) = \prod_{i\in I} 1_i$$

and

$$\left(\prod_{i\in I} a_i^{-1}\right)\left(\prod_{i\in I} a_i\right) = \prod_{i\in I}(a_i^{-1} \star_i a_i) = \prod_{i\in I} 1_i.$$

Thus, $\prod a_i^{-1}$ is the inverse of $\prod a_i$. We conclude that G is a group with inverse and identity as described above. ■

Exercise 5.1.16

State and prove the generalization of Proposition 2 to arbitrary direct products.

Solution. Proposition 2 stated for direct arbitrary groups is as follows: if $G = \prod_{i\in I} G_i$ is a direct product of groups, then

1. Fix each i with j-th coordinate where $j \neq i$ is the identity element of G_j for $1 \le j \le n$. Then

$$\widehat{G}_i = \{(1, 1, \dots, 1, g_i, 1, \dots) \mid g_i \in G_i\}$$

is a normal subgroup of G that is isomorphic to G_i, or that

$$G/\widehat{G}_i \cong \prod_{j\neq i} G_j.$$

2. For fixed i, define $\pi_i : G \to G_i$ by $\pi_i((g_j)_{j\in I}) = g_i$. Then π_i is a surjective homomorphism with kernel $\widehat{G}_i$, so $G/\widehat{G}_i \cong G_i$.
3. If $x \in G_i$ and $y \in G_j$ for $i \neq j$, then x and y commute in G.

The proof is similar to the proof of Proposition 2 in the text.

1. It is clear that $\widehat{G}_i$ is a subgroup of G. Moreover, for any $\prod_{j\in I} g_j \in G$ and any $h = \prod_{j\neq i} 1_j h_i \in \widehat{G}_i$, we have
$$\left(\prod_{j\in I} g_j\right) h \left(\prod_{j\in I} g_j\right)^{-1} = \prod_{j\neq i} 1_j (g_i h_i g_i^{-1}) \in \widehat{G}_i$$
since $g_i h_i g_i^{-1} \in G_i$. Thus, $\widehat{G}_i$ is normal in G. Consider the map $\varphi : G_i \to \widehat{G}_i$ defined by $\varphi(g_i) = \prod_{j\neq i} 1_j g_i$. It is clear that φ is a homomorphism. Moreover, if $\varphi(g_i) = \prod_{j\neq i} 1_j g_i = \prod_{j\neq i} 1_j$, then g_i is the identity element of G_i, so φ is injective. Finally, for any $\prod_{j\neq i} 1_j h_i \in \widehat{G}_i$, we have $\varphi(h_i) = \prod_{j\neq i} 1_j h_i$, so φ is surjective. Thus, $\widehat{G}_i$ is isomorphic to G_i.

   Consider the map
$$\psi : G \to \prod_{j\neq i} G_j \text{ defined by } \psi\left(\prod_{j\in I} g_j\right) = \prod_{j\neq i} g_j.$$
It is clear that ψ is a homomorphism. Moreover, the kernel of ψ is $\widehat{G}_i$, so by the First Isomorphism Theorem, we have
$$G/\widehat{G}_i \cong \prod_{j\neq i} G_j.$$

2. For fixed i, it is clear that π_i is a homomorphism. Moreover, π_i is surjective since for any $g_i \in G_i$, we have $\pi_i(\prod_{j\in I} 1_j g_i) = g_i$. Finally, the kernel of π_i is $\widehat{G}_i$, so by the First Isomorphism Theorem, we have
$$G/\widehat{G}_i \cong G_i.$$

3. Let $x = \prod_{j\neq i} 1_j x_i$ and $y = \prod_{j\neq k} 1_j y_k$ for some $i \neq k$. Then
$$xy = \prod_{j\neq i,k} 1_j x_i y_k = \prod_{j\neq i,k} 1_j y_k x_i = yx$$
as desired.

■

Exercise 5.1.17

Let I be any nonempty index set and let G_i be a group for each $i \in I$. The *restricted direct product* or *direct sum* of the groups G_i is the set of elements of the direct product which are the identity in all but finitely many components, that is, the set of all elements $\prod a_i \in \prod_{i\in I} G_i$ such that $a_i = 1_i$ for all but a finite number of $i \in I$.

(a) Prove that the restricted direct product is a subgroup of the direct product.

(b) Prove that the restricted direct product is normal in the direct product.

(c) Let $I = \mathbb{Z}^+$ and let p_i be the ith integer prime. Show that if $G_i = \mathbb{Z}/p_i\mathbb{Z}$ for all $i \in \mathbb{Z}^+$, then every element of the restricted direct product of the G_i's has finite order but $\prod_{i\in\mathbb{Z}^+} G_i$ has elements of infinite order. Show that in this example the restricted direct product is the torsion subgroup of the direct product (cf. Exercise 2.1.16).

Solution.

(a) Let $G = \prod_{i\in I} G_i$ and let H be the restricted direct product of the G_i's. It is clear that H is a subset of G. Moreover, if $\prod a_i$ and $\prod b_i$ are elements of H, then there are only finitely many i such that $a_i \neq 1_i$ and only finitely many i such that $b_i \neq 1_i$. Thus, there are only finitely many i such that $a_i b_i \neq 1_i$, so $\prod a_i \prod b_i = \prod(a_i b_i)$ is an element of H. Finally, if $\prod a_i$ is an element of H, then there are only finitely many i such that $a_i \neq 1_i$, so there are only finitely many i such that $a_i^{-1} \neq 1_i$, so $\prod a_i^{-1}$ is an element of H. Thus, $H \leq G$.

(b) Let $G = \prod_{i\in I} G_i$ and let H be the restricted direct product of the G_i's. Let $\prod a_i$ be an element of G and let $\prod b_i$ be an element of H. Then there are only finitely many i such that $b_i \neq 1_i$, so there are only finitely many i such that $a_i b_i a_i^{-1} \neq 1_i$, so $\prod a_i \prod b_i \prod a_i^{-1}$ is an element of H. Thus, $H \trianglelefteq G$.

(c) Let $G = \prod_{i \in \mathbb{Z}^+} G_i$ where $G_i = \mathbb{Z}/p_i\mathbb{Z}$ for all i. Let H be the restricted direct product of the G_i's. If $\prod a_i$ is an element of H, then there are only finitely many i such that $a_i \neq 1_i$, so there are only finitely many i such that a_i has order greater than 1, so $\prod a_i$ has finite order. On the other hand, if $\prod a_i$ is an element of G such that a_i is a generator of G_i for all i, then $\prod a_i$ has infinite order since for any positive integer n, we have $(\prod a_i)^n = \prod a_i^n$, and there are infinitely many i such that $a_i^n \neq 1_i$.

To show $H = \mathrm{Tor}(G)$, we note that $H \subseteq \mathrm{Tor}(G)$ since every element of H has finite order. Conversely, if $\prod a_i$ is an element of $\mathrm{Tor}(G)$, then there is some positive integer n such that $(\prod a_i)^n = \prod a_i^n = \prod 1_i$, so $a_i^n = 1_i$ for all i. Since G_i is cyclic of prime order, this implies that a_i is either the identity or a generator of G_i. If there are infinitely many i such that a_i is a generator of G_i, then $\prod a_i$ has infinite order, which is a contradiction. Thus, there are only finitely many i such that a_i is a generator of G_i, so there are only finitely many i such that $a_i \neq 1_i$, so $\prod a_i$ is an element of H. Therefore, $\mathrm{Tor}(G) \subseteq H$, and we conclude that $\mathrm{Tor}(G) = H$. ■

Exercise 5.1.18

In each of (a) to (e) give an example of a group with the specified properties:

(a) an infinite group in which every element has order 1 or 2

(b) an infinite group in which every element has finite order but for each positive integer n there is an element of order n

(c) a group with an element of infinite order and an element of order 2

(d) a group G such that every finite group is isomorphic to some subgroup of G

(e) a nontrivial group G such that $G \cong G \times G$.

Solution.

(a) Consider the infinite direct product $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \cdots$. Every element of this group is a tuple of 0's and 1's, and the order of each element is either 1 (if the tuple is the identity) or 2 (if the tuple has at least one 1).

(b) Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \cdots$. Every element of G has finite order since it is a tuple of elements from finite cyclic groups. Moreover, for each positive integer n, the element that is the identity in all components except the n-th component, which is a generator of $\mathbb{Z}/n\mathbb{Z}$, has order n.

(c) Consider the group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$. The element $(1, 0)$ has order 2, and the element $(0, 1)$ has infinite order.

(d) By Cayley's Theorem, every finite group is isomorphic to a subgroup of some symmetric group S_n. We may then consider the group $G = S_1 \times S_2 \times S_3 \times \cdots$. For any finite group H, there is some n such that H is isomorphic to a subgroup of S_n, so H is isomorphic to a subgroup of G.

(e) Let $G = \mathbb{Z} \times \mathbb{Z} \times \cdots$. Consider the map $\varphi : G \to G \times G$ defined by

$$\varphi((a_1, a_2, a_3, \ldots)) = ((a_1, a_3, a_5, \ldots), (a_2, a_4, a_6, \ldots)).$$

It is clear that φ is a homomorphism. Moreover, if $\varphi((a_1, a_2, a_3, \ldots)) = ((0, 0, 0, \ldots), (0, 0, 0, \ldots))$, then $a_i = 0$ for all i, so φ is injective. Finally, let $((b_1, b_2, b_3, \ldots), (c_1, c_2, c_3, \ldots))$ be an element of $G \times G$. Then $\varphi((b_1, c_1, b_2, c_2, b_3, c_3, \ldots)) = ((b_1, b_2, b_3, \ldots), (c_1, c_2, c_3, \ldots))$, so φ is surjective. Thus, φ is an isomorphism and $G \cong G \times G$. ■

5.2 The Fundamental Theorem of Finitely Generated Abelian Groups

Exercise 5.2.1

In each parts (a) to (e), give the number of nonisomorphic abelian groups of the specified order—do not list the groups:
(a) order 100, (b) order 576, (c) order 1155, (d) order 42875, (e) order 2704.

Solution.

(a) $100 = 2^2 \cdot 5^2$; 2 partitions of 2, hence $2 \cdot 2 = 4$ nonisomorphic abelian groups of order 100.
(b) $576 = 2^6 \cdot 3^2$; 11 partitions of 6, hence $11 \cdot 2 = 22$ nonisomorphic abelian groups of order 576.
(c) $1155 = 3 \cdot 5 \cdot 7 \cdot 11$. Only one abelian group of order 1155.
(d) $42875 = 5^3 \cdot 7^3$; 3 partitions of 3, hence $3 \cdot 3 = 9$ nonisomorphic abelian groups of order 42875.
(e) $2704 = 2^4 \cdot 13^2$; 5 partitions of 4, hence $5 \cdot 2 = 10$ nonisomorphic abelian groups of order 2704. ■

Exercise 5.2.2

In each of parts (a) to (e), give the lists of invariant factors for all abelian groups of the specified order:
(a) group 270, (b) order 9801, (c) order 320, (d) order 105, (e) order 44100.

Solution.

(a) $270 = 2 \cdot 3^3 \cdot 5$. The invariant factors are $\{270\}, \{90, 3\}, \{30, 3, 3\}$.
(b) $9801 = 3^4 \cdot 11^2$. The invariant factors are $\{9801\}, \{891, 11\}, \{3267, 3\}, \{297, 33\}, \{1089, 9\}, \{99, 99\}, \{1089, 3, 3\}, \{99, 33, 3\}, \{363, 3, 3, 3\}, \{33, 33, 3, 3\}$.
(c) $320 = 2^6 \cdot 5$. The invariant factors are $\{320\}, \{160, 2\}, \{80, 4\}, \{80, 2, 2\}, \{40, 8\}, \{40, 4, 2\}, \{40, 2, 2, 2\}, \{20, 4, 4\}, \{20, 4, 2, 2\}, \{20, 2, 2, 2, 2\}, \{10, 2, 2, 2, 2, 2\}$.
(d) $105 = 3 \cdot 5 \cdot 7$. The invariant factors are $\{105\}$.
(e) $44100 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2$. The invariant factors are $\{44100\}, \{22050, 2\}, \{14700, 3\}, \{8820, 5\}, \{6300, 7\}, \{4410, 10\}, \{3150, 14\}, \{2205, 15\}, \{1800, 21\}, \{1260, 35\}, \{1470, 30\}, \{1050, 42\}, \{630, 70\}, \{420, 105\}, \{210, 210\}$. ■

Exercise 5.2.3

In each of the parts (a) to (e), give the lists of elementary divisors for all abelian groups of the specified order, and then match each list with the corresponding list of invariant factors found in the preceding exercise:
(a) group 270, (b) order 9801, (c) order 320, (d) order 105, (e) order 44100.

Solution.

(a) $\{2, 27, 5\}, \{2, 9, 5, 3\}, \{2, 3, 5, 3, 3\}$.
(b) $\{81, 121\}, \{81, 11, 11\}, \{27, 121, 3\}, \{27, 11, 3, 11\}, \{9, 121, 9\}, \{9, 11, 9, 11\}, \{9, 121, 3, 3\}, \{9, 11, 3, 11, 3\}, \{3, 121, 3, 3, 3\}, \{3, 11, 3, 11, 3, 3\}$.
(c) $\{64, 5\}, \{32, 5, 2\}, \{16, 5, 4\}, \{16, 5, 2, 2\}, \{8, 5, 8\}, \{8, 5, 4, 2\}, \{8, 5, 2, 2, 2\}, \{4, 5, 4, 4\}, \{4, 5, 4, 2, 2\}, \{4, 5, 2, 2, 2, 2\}, \{2, 5, 2, 2, 2, 2, 2\}$.
(d) $\{3, 5, 7\}$.
(e) $\{4, 9, 25, 49\}, \{2, 9, 25, 49, 2\}, \{4, 3, 25, 49, 3\}, \{4, 9, 5, 49, 5\}, \{4, 9, 25, 7, 7\}, \{2, 3, 25, 49, 2, 3\}, \{2, 9, 5, 49, 2, 5\}, \{2, 9, 25, 7, 2, 7\}, \{4, 3, 5, 49, 3, 5\}, \{2, 3, 25, 49, 2, 3\}, \{2, 9, 5, 49, 2, 7\}, \{4, 3, 25, 7, 3, 7\}, \{2, 3, 5, 49, 2, 3, 5\}, \{2, 3, 25, 7, 2, 3, 7\}, \{4, 3, 5, 7, 3, 5, 7\}, \{2, 3, 5, 7, 2, 3, 5, 7\}$. ■

Exercise 5.2.4

In each of parts (a) to (d), determine which pairs of abelian groups listed are isomorphic (here the expression $\{a_1, a_2, \dots, a_k\}$ denotes the abelian group $Z_{a_1} \times Z_{a_2} \times \cdots \times Z_{a_k}$).

(a) $\{4,9\}$, $\{6,6\}$, $\{8,3\}$, $\{9,4\}$, $\{6,4\}$, $\{64\}$.

(b) $\{2^2, 2\cdot 3^2\}$, $\{2^2\cdot 3, 2\cdot 3\}$, $\{2^3\cdot 3^2\}$, $\{2^2\cdot 3^2, 2\}$.

(c) $\{5^2\cdot 7^2,\ 3^2\cdot 5\cdot 7\}$, $\{3^2\cdot 5^2\cdot 7,\ 5\cdot 7^2\}$, $\{3\cdot 5^2,\ 7^2,\ 3\cdot 5\cdot 7\}$, $\{5^2\cdot 7,\ 3^2\cdot 5,\ 7^2\}$.

(d) $\{2^2\cdot 5\cdot 7,\ 2^3\cdot 5^3,\ 2\cdot 5^2\}$, $\{2^3\cdot 5^3\cdot 7,\ 2^3\cdot 5^3\}$, $\{2^2,\ 2\cdot 7,\ 2^3,\ 5^3,\ 5^3\}$, $\{2\cdot 5^3,\ 2^2\cdot 5^3,\ 2^3,\ 7\}$.

Solution.

(a) $\{4,9\}$ and $\{9,4\}$ have the same elementary divisors, so they are isomorphic. In order, the other four groups have elementary divisors $\{3,2,3,2\}$, $\{8,3\}$, $\{3,2,4\}$, and $\{64\}$, so they are not isomorphic to each other or to $\{4,9\}$.

(b) $\{2^2, 2\cdot 3^2\}$ and $\{2^2\cdot 3^2, 2\}$ have the same elementary divisors, so they are isomorphic. In order, the other two groups have elementary divisors $\{2^2\cdot 3, 2\cdot 3\}$ and $\{2^3\cdot 3^2\}$, so they are not isomorphic to each other or to $\{2^2, 2\cdot 3^2\}$.

(c) The first, second, and fourth groups have the same elementary divisors $\{5^2, 7^2, 3^2, 5, 7\}$, so they are isomorphic. The third group has elementary divisors $\{3, 5^2, 7^2, 3, 5, 7\}$, so it is not isomorphic to the other three groups.

(d) The third and fourth groups have the elementary divisors $\{2, 125, 4, 125, 8, 7\}$, so they are isomorphic. ■

Exercise 5.2.5

Let G be a finite abelian group of type $(n_1, n_2, \dots, n_t)$. Prove that G contains an element of order m if and only if $m \mid n_1$. Deduce that G is of exponent n_1.

Solution.

($\Rightarrow$) Let $g = (g_1, g_2, \dots, g_t) \in G$ have order m. Then $m = \operatorname{lcm}(|g_1|, |g_2|, \dots, |g_t|)$, so $|g_i| \mid m$ for all i. Since G is of type $(n_1, n_2, \dots, n_t)$, we have $|g_i| \mid n_i$ for all i, so m divides $\operatorname{lcm}(n_1, n_2, \dots, n_t) = n_1$.

($\Leftarrow$) If $m \mid n_1$, then there exists an element of order m in the cyclic subgroup of order n_1.

Since G contains an element of order n_1, the exponent of G is at least n_1. On the other hand, if $g \in G$, then $|g| \mid n_1$, so the exponent of G divides n_1. Thus, the exponent of G is exactly n_1. ■

Exercise 5.2.6

Prove that any finite group has a finite exponent. Give an example of an infinite group with finite exponent. Does a finite group of exponent m always contain an element of order m?

Solution. Let G be a finite group. Then there are only finitely many elements of G, so there are only finitely many orders of elements in G. Let m be the least common multiple of the orders of all elements of G. Then for any $g \in G$, we have $g^m = e$, so G has exponent dividing m.

An example of an infinite group with finite exponent is the infinite direct product $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \cdots$. Every element of this group has order 1 or 2, so the exponent of this group is 2.

A finite group of exponent m does not always contain an element of order m: consider S_3, which has exponent 6 but does not contain an element of order 6. ■

Exercise 5.2.7

Let p be a prime and let $A = \langle x_1 \rangle \times \langle x_2 \rangle \times \cdots \times \langle x_n \rangle$ be an abelian p-group, where $|x_i| = p^{\alpha_i} > 1$ for all i. Define the p^{th}*-power map*

$$\varphi : A \to A \quad \text{by} \quad \varphi : x \mapsto x^p.$$

(a) Prove that φ is a homomorphism.
(b) Describe the image and kernel of φ in terms of the given generators.
(c) Prove both $\ker\varphi$ and $A/\operatorname{im}\varphi$ have rank n (i.e., have the same rank as A) and prove these groups are both isomorphic to the elementary abelian group, E_{p^n}, of order p^n.

Solution.

(a) Let $x = (x_1^{s_1}, x_2^{s_2}, \ldots, x_n^{s_n})$ and $y = (x_1^{t_1}, x_2^{t_2}, \ldots, x_n^{t_n})$ be elements of A. Then

$$\begin{aligned}
\varphi(xy) &= \varphi((x_1^{s_1+t_1}, x_2^{s_2+t_2}, \ldots, x_n^{s_n+t_n})) \\
&= (x_1^{p(s_1+t_1)}, x_2^{p(s_2+t_2)}, \ldots, x_n^{p(s_n+t_n)}) \\
&= (x_1^{ps_1}x_1^{pt_1}, x_2^{ps_2}x_2^{pt_2}, \ldots, x_n^{ps_n}x_n^{pt_n}) \\
&= (x_1^{ps_1}, x_2^{ps_2}, \ldots, x_n^{ps_n})(x_1^{pt_1}, x_2^{pt_2}, \ldots, x_n^{pt_n}) \\
&= \varphi(x)\varphi(y)
\end{aligned}$$

as desired.

(b) φ collects all the p-powers, so $\operatorname{im}\varphi = \langle x_1^p \rangle \times \langle x_2^p \rangle \times \cdots \times \langle x_n^p \rangle$. Moreover, $\ker\varphi$ consists of all elements $(x_1^{s_1}, x_2^{s_2}, \ldots, x_n^{s_n})$ such that $x_1^{ps_1} = x_2^{ps_2} = \cdots = x_n^{ps_n} = e$, which implies that ps_i is a multiple of p^{α_i} for all i, so s_i is a multiple of $p^{\alpha_i - 1}$ for all i. Thus,

$$\ker\varphi = \left\langle x_1^{p^{\alpha_1 - 1}} \right\rangle \times \left\langle x_2^{p^{\alpha_2 - 1}} \right\rangle \times \cdots \times \left\langle x_n^{p^{\alpha_n - 1}} \right\rangle.$$

(c) Note that $x_i^{p^{\alpha_i - 1}}$ has order p for all i, so $\ker\varphi$ is an elementary abelian group of order p^n. Moreover, note that each $\langle x_i \rangle$ is cyclic, hence every group is normal. Examining $\langle x_i \rangle / \langle x_i^p \rangle$, we see that $\langle x_i \rangle / \langle x_i^p \rangle$ is an elementary abelian group of order p. Since $A/\operatorname{im}\varphi$ is the direct product of the $\langle x_i \rangle / \langle x_i^p \rangle$, we conclude that $A/\operatorname{im}\varphi$ is an elementary abelian group of order p^n, by Exercise 5.1.14. ■

Exercise 5.2.8

Let A be a finite abelian group (written multiplicatively) and let p be a prime. Let

$$A^p = \{a^p \mid a \in A\} \qquad \text{and} \qquad A_p = \{x \mid x^p = 1\}$$

(so A^p and A_p are the image and kernel of the p^{th}-power map, respectively).

(a) Prove that $A/A^p \cong A_p$. [Show that they are both elementary abelian and they have the same order.]
(b) Prove that the number of subgroups of A of order p equals the number of subgroups of A of index p. [Reduce to the case where A is an elementary abelian p-group.]

Solution.

(a) By Theorem 5.5, we may write A as a direct product:

$$A \cong A_{p_1} \times A_{p_2} \times \cdots \times A_{p_k},$$

where each A_{p_i} is a p_i-group for distinct primes $p_1, p_2, \ldots, p_k$. There are two cases:

- If $p \neq p_i$ for all i, then then $(p, p_i) = 1$ for all i, so the p-th power map is an automorphism of each A_{p_i}, so $A^p = A$ and $A_p = \{1\}$, so $A/A^p \cong A_p$.

- If $p = p_i$ for some i, then it is an automorphism for all A_j for $j \neq i$, while the case for A_{p_i} is handled by Exercise 5.2.7. Thus,
$$A^p = A_{p_1} \times \cdots \times A_{p_{i-1}} \times A_{p_i}^p \times A_{p_{i+1}} \times \cdots \times A_{p_k}$$
and
$$A_p = \{1\} \times \cdots \times \{1\} \times (A_{p_i})_p \times \{1\} \times \cdots \times \{1\}.$$

We may again use Exercise 5.1.14 along with the fact that $A_{p_i}/A_{p_i}^p \cong A_{p_i}^p$ to conclude that $A/A^p \cong A_p$.

(b) Let $H \leq A$ such that $|A : H| = p$. Let $a \in A$. Then $aH \in A/H$. Since $|A : H| = p$, then $|aH| = 1$ or p. If $|aH| = 1$, then $a \in H$. If $|aH| = p$, then $a^pH = H$, or $a^p \in H$. In either case, we have $a^p \in H$, so $A^p \subseteq H$. By the Lattice Isomorphism Theorem, then subgroups of A containing A^p are in bijection with subgroups of A/A^p. By the previous part, $A/A^p \cong A_p$, so the number of index p subgroups of A equals the number of index p subgroups of A_p.

Now let $K \leq A$ where $|K| = p$. For any $k \in K$, we know $k^p = 1$, so $K \subseteq A_p$. Then subgroups of order p in A are exactly the subgroups of order p in A_p, since $A_p \leq A$. Thus, the number of subgroups of order p in A equals the number of subgroups of order p in A_p.

Note that A_p is an elementary abelian p-group, so it is isomorphic to E_{p^n} for some n. By Exercise 5.1.11, the number of subgroups of order p in E_{p^n} is $\frac{p^n-1}{p-1}$. To see that the number of subgroups of index p in E_{p^n} is also $\frac{p^n-1}{p-1}$, we note that there are $p^n - 1$ surjective homomorphisms from E_{p^n} to E_p, and each subgroup of index p is the kernel of exactly $p-1$ such homomorphisms, so there are $\frac{p^n-1}{p-1}$ subgroups of index p in E_{p^n}. ■

Exercise 5.2.9

Let $A = Z_{60} \times Z_{45} \times Z_{12} \times Z_{36}$. Find the number of elements of order 2 and the number of subgroups of index 2 in A.

Solution. Note that
- $Z_{60} \cong Z_4 \times Z_3 \times Z_5$,
- $Z_{45} \cong Z_9 \times Z_5$,
- $Z_{12} \cong Z_4 \times Z_3$, and
- $Z_{36} \cong Z_4 \times Z_9$.

Thus, the 2-power part of A is $Z_4 \times Z_4 \times Z_4$. Elements of order 2 in A are elements of the 2-power part as well as trivial parts of the other groups. The 2-power map sends (a, b, c) to $(2a, 2b, 2c)$, so the kernel of the 2-power map is $\{0, 2\} \times \{0, 2\} \times \{0, 2\}$, which has 8 elements, 7 of which have order 2. Thus, there are 7 elements of order 2 in A. ■

Exercise 5.2.10

Let n and k be positive integers and let A be the free abelian group of rank n (written additively). Prove that A/kA is isomorphic to the direct product of n copies of $\mathbb{Z}/k\mathbb{Z}$ (here, $kA = \{ka \mid a \in A\}$). [See Exercise 5.1.14.]

Solution. It is easy to see that $kA = k\mathbb{Z} \times \cdots \times k\mathbb{Z}$. Since $\mathbb{Z}$ is abelian, then $k\mathbb{Z} \trianglelefteq \mathbb{Z}$. By Exercise 5.1.14, we have $A/kA \cong \mathbb{Z}/k\mathbb{Z} \times \cdots \times \mathbb{Z}/k\mathbb{Z}$. ■

Exercise 5.2.11

Let G be a nontrivial, finite abelian group of rank t.

(a) Prove that the rank of G equals the maximum of the ranks of its Sylow subgroups.

(b) Prove that G can be generated by t elements, but no subset with fewer than t elements generates G. (One way of doing this is by using part(a) together with Exercise 5.2.7.)

Solution.

(a) Let G be of type $(n_1, n_2, \ldots, n_t)$, and consider the invariant factor decomposition of G:

$$G \cong Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_t}.$$

Let p be a prime dividing the smallest factor n_t. Then the Sylow p-subgroup of G is isomorphic to $Z_{p^{\alpha_1}} \times Z_{p^{\alpha_2}} \times \cdots \times Z_{p^{\alpha_t}}$, where α_i is the largest power of p dividing n_i. Since $\alpha_t > 0$, the Sylow p-subgroup has rank t. On the other hand, if q is any prime, then for any $Q \in \mathrm{Syl}_q(G)$, we have

$$Q \cong Z_{q^{\beta_1}} \times Z_{q^{\beta_2}} \times \cdots \times Z_{q^{\beta_t}},$$

where $\beta_i \mid n_i$ for all i with some potentially trivial factors, so the rank of Q is at most t. Thus, the rank of G equals the maximum of the ranks of its Sylow subgroups.

(b) It is clear that G is generated by t elements. Suppose $G = \langle g_1, g_2, \ldots, g_{t-1}\rangle$ and some $P \in \mathrm{Syl}_p(G)$, which has rank t. Projecting G onto P, we see that $P = \langle \pi(g_1), \pi(g_2), \ldots, \pi(g_{t-1})\rangle$. However, letting φ be the p-power map on P as defined in Exercise 5.2.7, we have that $P/\operatorname{im}\varphi \cong E_{p^t}$, so $P/\operatorname{im}\varphi$ has rank t. Since $\pi(g_1), \pi(g_2), \ldots, \pi(g_{t-1})$ generate $P/\operatorname{im}\varphi$, we have a contradiction. Thus, no subset with fewer than t elements generates G. ■

(∗) Exercise 5.2.12

Let n and m be positive integers with $d = (n, m)$. Let $Z_n = \langle x\rangle$ and $Z_m = \langle y\rangle$. Let A be the central product of $\langle x\rangle$ and $\langle y\rangle$ with an element of order d identified, which has presentation

$$\langle x, y \mid x^n = y^m = 1, xy = yx, x^{n/d} = y^{m/d}\rangle.$$

Describe A as a direct product of two cyclic groups.

Note: While the problem specifically calls for two direct products, the proof shows that for any general n, m, A is actually cyclic with order $Z_{nm/d}$, so the second direct product is trivial. I'm not sure if this is a typo, but just know that the problem will have a trivial direct product in the end.

Solution. Since we have an element of order d identified, it is easy to see that $|A| = nm/d$. By Exercise 5.2.5, we know that $\exp(Z_n \times Z_m) = \operatorname{lcm}(n, m) = nm/d$. Note that $|x| = n$ and $|y| = m$ in A still, for otherwise there would be a contradiction to the order of A as we would not be able to account for all the elements of A.

Again, by Exercise 5.2.5, we know that A contains an element of order n iff $n \mid \exp(A)$, and A contains an element of order m iff $m \mid \exp(A)$. Since $\exp(A) \mid |A| = nm/d$, we have that $\exp(A)$ is a common multiple of n and m, so $\exp(A)$ is a multiple of $\operatorname{lcm}(n, m) = nm/d$, hence $\operatorname{lcm}(n, m) \mid \exp(A)$. On the other hand, $\exp(A) \mid |A| = nm/d$ since the order of any element divides the order of the group, so $\exp(A) \mid \operatorname{lcm}(n, m)$. Thus, $\exp(A) = \operatorname{lcm}(n, m)$. We now conclude that A is cyclic, so $A \cong Z_{nm/d} \times Z_1$. ■

Exercise 5.2.13

Let $A = \langle x_1 \rangle \times \langle x_2 \rangle \times \cdots \times \langle x_r \rangle$ be a finite abelian group with $|x_i| = n_i$ for $1 \le i \le r$. Find a presentation for A. Prove that if G is any group containing commuting elements $g_1, \dots, g_r$ such that $g_i^{n_i} = 1$ for $1 \le i \le r$, then there is a unique homomorphism from A to G which sends x_i to g_i for all i.

Solution. A presentation for A is given by

$$\langle x_1, x_2, \dots, x_r \mid x_i^{n_i} = 1 \text{ for } 1 \le i \le r, x_i x_j = x_j x_i \text{ for } 1 \le i < j \le r \rangle.$$

Now, observe that every element of A is of the form $x_1^{s_1} x_2^{s_2} \cdots x_r^{s_r}$ for some integers $s_1, s_2, \dots, s_r$. Define $\varphi : A \to G$ by $\varphi(x_1^{s_1} x_2^{s_2} \cdots x_r^{s_r}) = g_1^{s_1} g_2^{s_2} \cdots g_r^{s_r}$. Since the g_i commute, φ is well-defined. Moreover, φ is a homomorphism since

$$\begin{aligned} \varphi((x_1^{s_1} x_2^{s_2} \cdots x_r^{s_r})(x_1^{t_1} x_2^{t_2} \cdots x_r^{t_r})) &= \varphi(x_1^{s_1+t_1} x_2^{s_2+t_2} \cdots x_r^{s_r+t_r}) \\ &= g_1^{s_1+t_1} g_2^{s_2+t_2} \cdots g_r^{s_r+t_r} \\ &= (g_1^{s_1} g_2^{s_2} \cdots g_r^{s_r})(g_1^{t_1} g_2^{t_2} \cdots g_r^{t_r}) \\ &= \varphi(x_1^{s_1} x_2^{s_2} \cdots x_r^{s_r}) \varphi(x_1^{t_1} x_2^{t_2} \cdots x_r^{t_r}) \end{aligned}$$

for all integers $s_1, s_2, \dots, s_r$ and $t_1, t_2, \dots, t_r$. Finally, $\varphi(x_i) = g_i$ for all i by definition. To see that φ is the unique homomorphism from A to G sending x_i to g_i for all i, let $\psi : A \to G$ be any such homomorphism. Then $\psi(x_1^{s_1} x_2^{s_2} \cdots x_r^{s_r}) = \psi(x_1)^{s_1} \psi(x_2)^{s_2} \cdots \psi(x_r)^{s_r} = g_1^{s_1} g_2^{s_2} \cdots g_r^{s_r}$ for all integers $s_1, s_2, \dots, s_r$, so $\psi = \varphi$. ■

(∗) Exercise 5.2.14

For any group G define the *dual group* of G (denoted $\widehat{G}$) to be the set of all homomorphisms from G into the multiplicative group of roots of unity in $\mathbb{C}$. Define a group operation in $\widehat{G}$ by pointwise multiplication of functions: if χ, ψ are homomorphisms from G into the group of roots of unity then $\chi\psi$ is the homomorphism given by $(\chi\psi)(g) = \chi(g)\psi(g)$ for all $g \in G$, where the latter multiplication takes place in $\mathbb{C}$.

(a) Show that this operation on $\widehat{G}$ makes $\widehat{G}$ into an abelian group. [Show that the identity is the map $g \mapsto 1$ for all $g \in G$ and the inverse of $\chi \in \widehat{G}$ is the map $g \mapsto \chi(g)^{-1}$.]

(b) If G is a finite abelian group, prove that $\widehat{G} \cong G$. [Write G as $\langle x_1 \rangle \times \cdots \times \langle x_r \rangle$ and if $n_i = |x_i|$ define χ_i to be the homomorphism which sends x_i to $e^{2\pi i/n_i}$ and sends x_j to 1, for all $j \ne i$. Prove χ_i has order n_i in $\widehat{G}$ and $\widehat{G} = \langle \chi_1 \rangle \times \cdots \times \langle \chi_r \rangle$.]

Solution.

(a) Multiplication in $\mathbb{C}$ is already associative and commutative, so pointwise multiplication of functions is also associative and commutative. The map $\epsilon : G \to \mathbb{C}$ given by $\epsilon(g) = 1$ for all $g \in G$ is a homomorphism, since $\epsilon(gh) = 1 = 1 \cdot 1 = \epsilon(g)\epsilon(h)$ for all $g, h \in G$, and 1 is a root of unity. Moreover, for any $\chi \in \widehat{G}$, then $(\chi\epsilon)(g) = \chi(g)\epsilon(g) = \chi(g) \cdot 1 = \chi(g)$ for all $g \in G$. Thus, ϵ is the identity element of $\widehat{G}$. Finally, if $\chi \in \widehat{G}$, then the map $\chi^{-1} : G \to \mathbb{C}$ given by $\chi^{-1}(g) = \chi(g)^{-1}$ is a homomorphism, since $\chi^{-1}(gh) = \chi(gh)^{-1} = (\chi(g)\chi(h))^{-1} = \chi(g)^{-1}\chi(h)^{-1} = \chi^{-1}(g)\chi^{-1}(h)$ for all $g, h \in G$, and $\chi(g)^{-1}$ is a root of unity since $\chi(g)$ is a root of unity. Moreover, $\chi\chi^{-1} = \epsilon$, so χ^{-1} is the inverse of χ in $\widehat{G}$.

(b) To see that χ_i has order n_i in $\widehat{G}$, let $g = x_1^{s_1} \cdots x_r^{s_r} \in G$. Then

$$\chi_i(g) = \chi_i(x_1^{s_1} \cdots x_r^{s_r}) = \chi_i(x_1)^{s_1} \cdots \chi_i(x_r)^{s_r} = e^{2\pi i s_i/n_i},$$

where all terms $\chi_i(x_j)^{s_j}$ for $j \ne i$ are 1. Thus, $\chi_i^k(g) = e^{2\pi i k s_i/n_i}$ for all integers k, so $\chi_i^k = \epsilon$ if and only if $n_i \mid k$. Thus, χ_i has order n_i in $\widehat{G}$. Now, note by Exercise 5.2.13 that there exists a unique homomorphism $\varphi : G \to \widehat{G}$ such that $\varphi(x_i) = \chi_i$ for all i. We now show that φ is an isomorphism.

Let $g = x_1^{s_1} \cdots x_r^{s_r} \in G$ such that $\varphi(g) = \epsilon$. Then for any x_i, we have that

$$\varphi(g)(x_i) = (\chi_1^{s_1} \cdots \chi_r^{s_r})(x_i) = \chi_i^{s_i}(x_i) = e^{2\pi i s_i/n_i} = 1.$$

Thus, $n_i \mid s_i$ for all i, so $g = x_1^{s_1} \cdots x_r^{s_r} = 1$. Therefore, φ is injective. Similarly, for any $\chi \in \widehat{G}$, and $x_i^{n_i} = 1$ for some i, then $\chi(x_i)^{n_i} = \chi(x_i^{n_i}) = \chi(1) = 1$, so $\chi(x_i)$ is an n_i-th root of unity, so $\chi(x_i) = e^{2\pi i s_i/n_i}$ for some integer s_i. Thus, $\chi = \varphi(x_1^{s_1} \cdots x_r^{s_r})$, so φ is surjective. Therefore, φ is an isomorphism, so $\widehat{G} \cong G$. ■

Exercise 5.2.15

Let $G = \langle x \rangle \times \langle y \rangle$, where $|x| = 8$ and $|y| = 4$.

(a) Find all pairs a, b in G such that $G = \langle a \rangle \times \langle b \rangle$ (where a and b are expressed in terms of x and y).

(b) Let $H = \langle x^2 y, y^2 \rangle \cong Z_4 \times Z_2$. Prove that there are no elements a, b of G such that $G = \langle a \rangle \times \langle b \rangle$ and $H = \langle a^2 \rangle \times \langle b^2 \rangle$ (i.e., one cannot pick direct product generators for G in such a way that some powers of these are direct product generators for H).

Solution.

(a) From the description of G, we see that G is of type $(8, 4)$, so we have the following conditions for $G = \langle a \rangle \times \langle b \rangle$:
- $|a| = 8$ and $|b| = 4$,
- $\langle a \rangle \cap \langle b \rangle = 1$.

Let $a = x^s y^t$ and $b = x^u y^v$ for some integers s, t, u, v. Then $|a| = \operatorname{lcm}(8/\gcd(8, s), 4/\gcd(4, t))$ and $|b| = \operatorname{lcm}(8/\gcd(8, u), 4/\gcd(4, v))$. For $|a|$ to be 8, we must have $\gcd(8, s) = 1$, implying that s must be odd, while t may range freely. For $|b|$ to be 4, then $8/\gcd(8, u) \mid 4$ and $\max(8/\gcd(8, u), 4/\gcd(4, v)) = 4$, so we have the following cases:

(1) If $\gcd(8, u) = 8$, then $u = 0$ and v must be odd.
(2) If $\gcd(8, u) = 4$, then $u = 4$ and v must be odd.
(3) If $\gcd(8, u) = 2$, then $u = 2, 6$ and v can be free.

To have trivial intersection, we must have that $x^{ks} y^{kt} = x^{\ell u} y^{\ell v} = 1$, or $ks \equiv \ell u \bmod 8$ and $kt \equiv \ell v \bmod 4$ implies that $k \equiv 0 \bmod 8$ and $\ell \equiv 0 \bmod 4$, which is seen from $a^k = 1$ and $b^\ell = 1$ iff $8 \mid k$ and $4 \mid \ell$. Thus, we check the cases for $|b|$:

(1) $u = 0$ and v is odd: We have $ks \equiv 0 \bmod 8$ and $kt \equiv \ell v \bmod 4$. Since s is odd, then $k \equiv 0 \bmod 8$. Then $kt \equiv 0 \bmod 4$, hence $\ell v \equiv 0 \bmod 4$. Since v is odd, then $\ell \equiv 0 \bmod 4$, so we have trivial intersection.

(2) $u = 4$ and v is odd: We have $ks \equiv 4\ell \bmod 8$ and $kt \equiv \ell v \bmod 4$. Since s is odd, then $k \equiv 4\ell s^{-1} \bmod 8$. Note that $k \equiv 0 \bmod 4$, so $kt \equiv 0 \bmod 4$, hence $\ell v \equiv 0 \bmod 4$. Since v is odd, then $\ell \equiv 0 \bmod 4$, so we have trivial intersection.

(3) $u = 2$ and v free (the case $u = 6$ and v free is similar): We have $ks \equiv 2\ell \bmod 8$ and $kt \equiv \ell v \bmod 4$. Since s is odd, then $k \equiv 2\ell s^{-1} \bmod 8$. We know that intersection will be nontrivial when $k \not\equiv 0 \bmod 8$ and $\ell \not\equiv 0 \bmod 4$. By inspection, we see that $\ell = 2$ gives $k \equiv 4s^{-1} \equiv 4 \bmod 8$ and $4t \equiv 2v \bmod 4$, which has nontrivial solution when v is even. Inspecting odd v, we see that $k \equiv 2\ell s^{-1} \bmod 8$ reduced mod 4 gives $k \equiv 2\ell s^{-1} \bmod 4$. Then $2\ell s^{-1} \equiv \ell v \bmod 4$ implies that $\ell(2s^{-1} - v) \equiv 0 \bmod 4$. Since v is odd, then $2s^{-1} - v$ is odd, so $\ell \equiv 0 \bmod 4$, so we have trivial intersection.

Therefore, the pairs a, b such that $G = \langle a \rangle \times \langle b \rangle$ are given by
- $a = x^s y^t$ and $b = y^v$ for some odd integers s, v and any integer t,
- $a = x^s y^t$ and $b = x^4 y^v$ for some odd integers s, v and any integer t,
- $a = x^s y^t$ and $b = x^2 y^v$ for some odd integer s, any integer t, and any odd integer v, or
- $a = x^s y^t$ and $b = x^6 y^v$ for some odd integer s, any integer t, and any odd integer v.

(b) We begin with the elements of H:

$$H = \{1, x^2 y, x^4 y^2, x^6 y^3, y^2, x^2 y^3, x^4 y, x^6 y^2\}$$

Note that $H = \langle x^2 y \rangle \times \langle y^2 \rangle$, so it suffices to check whether or not a^2 generates $\langle x^2 y \rangle$ from the choices of a from part (a). Note that $a^2 = x^{2s} y^{2t}$, which implies that y will always have an even exponent. However, the only element with even exponent on y from $\langle x^2 y \rangle$ is with the term $x^4 y^2$, which would imply that $\langle x^4 y^2 \rangle = \langle x^2 y \rangle$, which is a contradiction since $x^2 y$ has order 4 while $x^4 y^2$ has order 2. Thus, there are no elements a, b of G such that $G = \langle a \rangle \times \langle b \rangle$ and $H = \langle a^2 \rangle \times \langle b^2 \rangle$. ■

Exercise 5.2.16

Prove that no finitely generated abelian group is divisible (cf. Exercise 2.4.19).

Solution. Let A be a finitely generated abelian group. By Theorem 5.5, we can write A as a direct product of cyclic groups:

$$A \cong \mathbb{Z}^r \times Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_t},$$

where $n_i > 1$ for all i and $r \geq 0$. For $r > 0$, observe that $\mathbb{Z}$ is not divisible, since there is no $x \in \mathbb{Z}$ such that $2x = 1$. For $r = 0$, observe that A is a finite abelian group. By Exercise 2.4.19, finite abelian groups are not divisible. Thus, no finitely generated abelian group is divisible. ∎